

KALVIN WENG

(607) – 484 – 7870 | kweng2@binghamton.edu | [linkedin.com/in/kalvin-weng](https://www.linkedin.com/in/kalvin-weng) | github.com/kweng2bing

EDUCATION

Binghamton University, SUNY | Watson College of Engineering and Applied Science

Binghamton, NY

Bachelor of Science in Computer Science & Mathematical Sciences Minor

Cumulative GPA: 3.88/4.00 | **Dean's List:** Fall 2022 – Fall 2025 | NYS Scholarship for Academic Excellence

Relevant Coursework: Deep Learning, Machine Learning, Natural Language Processing, Advanced OOP, Program Models for Emerging Platforms, Computer Networks, Operating Systems, Data Structure & Algorithms, Design & Analysis of Algorithm, Topology, Advanced Linear Algebra, Statistics, Probability Theory, Multivariable Calculus, Number Systems (Mathematical Proof)

PROFESSIONAL EXPERIENCE

NYC Department of Environmental Protection | Backend Engineer Intern | Queens, NY

June 2025 – August 2025

- Programmed and implemented REST API integration to consolidate over 20K user data from Learning Management System, maintaining consistency, data integrity and improving operational efficiency by reducing weeks of manual data entry to hours
- Architected a scalable infrastructure and modular codebase that streamlined developer onboarding and significantly accelerated future development cycles, laying the foundation for rapid feature expansion and long-term maintainability
- Enhanced API efficiency and data handling logic, reducing redundant calls resulting a 50 % boost in data processing speed

Horizon Media | Search Engine Marketing Intern | New York, NY

June 2024 – August 2024

- Conducted daily optimization of Google Ads and Bing Ads campaigns, leveraging KPI-driven insights to uphold budget discipline, enhance conversion performance, and proactively mitigate fluctuations in campaign efficiency
- Examined over 1,000 search keywords daily from Google Ads and Bing Ads to identify high-value search terms and negative search terms, optimizing paid search campaigns for superior targeting and increasing click-through rates
- Pinpointed emerging trends in anime-related advertisements and delivered a company-wide presentation to over 2000 people, detailing strategic advantages and actionable steps for Horizon Media to capitalize on anime-related marketing strategies

Google | Computer Science Summer Institute Intern | (Remote) New York, NY

July 2022 – August 2022

- Developed fully functional web applications using HTML, CSS, and JavaScript, applying core software engineering and programming concepts such as variables, data types, functions, and product management across three interactive projects
- Coordinated with team members to develop a collaborative final project that included a live demonstration of its features and benefits to Google employees and community leaders

PROJECTS

Author Style LLM Fine Tuning Project | PyTorch, HuggingFace Transformers, Google Colab

November 2025 – Present

- Fine-tuned a large language model on a single author's works, building data processing pipelines, adapting tokenizers, and applying parameter-efficient training methods while avoiding overfitting and preserving broad prompt-response generality
- Produced a comprehensive evaluation report integrating quantitative metrics (loss curves, perplexity) with qualitative analyses of prompt-response alignment, establishing a structured framework for assessing author-emulation performance

Search and Rescue Drone Simulator | AirSim, Flask, Unreal Engine

September 2025 – Present

- Engineered a modular drone simulation framework in Microsoft AirSim that captures real-time spatial data with LiDAR sensors, while enabling scalable sensor fusion pipelines and high-fidelity spatial awareness for search-and-rescue scenarios
- Built a live update pipeline connecting multi-agent drone LiDAR feeds to a Flask-based local host dashboard, delivering synchronized 3D environment mapping and immediate visualization of sensor performance during concurrent missions

Machine Learning-Enhanced Trading Portfolio | Scikit-learn, NumPy, Pandas, TensorFlow, Alpaca

December 2024 – Present

- Developed a machine learning-enhanced trading portfolio that leverages historical stock data and technical indicators to train machine learning models to predict stock price movements and make inform trading decisions

Computer Network Communication System | C++

March 2025 – May 2025

- Engineered a robust computer network communication system in C++ that simulates key networking protocols, including TCP and UDP, to facilitate reliable data transmission between multiple clients and a server
- Designed a web server and proxy server to transfer files, incorporating socket programming for reliable communication and multithreading for efficient handling of multiple client requests

WORK EXPERIENCE

NYC Department of Citywide Administrative Services | Customer Support & Monitor | New York, NY

July 2023 – January 2024

- Managed the administration of over 300 computer-based exams daily by coordinating candidate intake and swiftly resolving technical issues to minimize disruptions and ensure a seamless testing experience addressing technical issues promptly

TECHNICAL SKILLS

Languages: Python, C++, C, Java, LaTeX, SQL, JavaScript

Tools and Frameworks: Eclipse, Git, Github, Google Analytics, React, VS Code, JUnit, Vim